Section 2 | Retrieval-Ready Knowledge, Chunking & Metadata

DSBA 6171 — A3 Revised Knowledge Retrieval Architecture
Evidence-based comparison of two character-based chunking configurations using five documents and six unfiltered Top-5 retrieval runs.

1. Corpus, ingestion and retrieval setup

Five source documents were ingested into a local DuckDB-backed retrieval service: the CFPB adverse-action circular (DOC001), the current loan-approval escalation procedure (FIN-PROC-001 v2), and underwriting policies FIN-POL-001 v1, v2 and v3. Registry metadata identifies v1 and v2 as **SUPERSEDED** and v3 as **CURRENT**. The escalation procedure and CFPB circular are also marked CURRENT.

The n8n ingestion workflow builds a document manifest, sends it to the Python service for chunking and embedding, and previews chunk metadata. Retrieval uses the same question wording, **nomic-embed-text:latest** embedding model and **Top-K = 5** for both configurations. The saved comparison requests have no authority filter, so outdated documents remain eligible to appear.

2. Chunking configurations and document coverage

Configuration	Chunk size	Overlap	Total chunks
A — baseline	1,600 characters	200 characters	69
B — alternative	800 characters	100 characters	144

Document	A chunks	B chunks
CFPB adverse-action circular	16	33
Loan approval procedure	27	57
2024 superseded policy	8	17
2026 revised current policy	13	27
2026 superseded policy	5	10

Configuration B produces approximately **2.09 times** as many chunks as A. Its shorter passages can be more focused, but create more vectors to embed, store and search. A retains more surrounding text per result, potentially improving context continuity.

3. Metadata and provenance design

Each retrieved chunk carries a stable **chunk_id**, **doc_id**, **version**, **chunk_index**, **start_char**, **source_path**, **effective_date**, **authority_status**, **confidentiality**, the chunk text, and registry metadata. The registry additionally supplies title, owner/source, region and access group. This supports source tracing, version selection and access-policy design; access-group enforcement was not demonstrated by these retrieval tests.

For example, the top result for Question 3 under B is **FIN-PROC-001 v2**, authority CURRENT, confidentiality Internal, effective 2026-01-01, source *loan_approval_escalation_procedure.md*, chunk index 2, similarity 0.734524. These fields make the retrieved evidence auditable.

4. Representative retrieved chunks

CFPB specificity — A, Q2 rank 1 — DOC001 v1.0, CURRENT, chunk 9, similarity 0.78503

“6 Specificity is particularly important when creditors utilize complex algorithms. Consumers may not anticipate that certain data gathered outside of their application or credit file and fed into an algorithmic decision-making model may be a principal reason in a credit decision, particularly if 12 12 CFR Par…”

Escalation status — B, Q3 rank 3 — FIN-PROC-001 v2, CURRENT, chunk 12, similarity 0.714332

“ly these exact case-sensitive business values may be stored in `loan_applications.decision_status`: | Valid Status | Use | |---|---| | `Approved` | All automatic requirements are satisfied and no escalation condition exists | | `Denied` | An explicit policy rule prevents approval and the specific principal re…”

Outdated policy risk — A, Q1 rank 1 — FIN-POL-001 v1, SUPERSEDED, chunk 0, similarity 0.705913

“# Internal Personal Loan Underwriting Policy > **SYNTHETIC INTERNAL DOCUMENT - SUPERSEDED** > Historical policy retained for academic retrieval, audit, and version-control testing. > **Do not use this version for applications dated on or after January 1, 2026.** | Document Metadata | Value | |---|---| | Docum…”

5. Three-question retrieval evaluation

Q1. Which underwriting policy version is currently authoritative?

Measure	Baseline A	Alternative B
Rank-1 result	FIN-POL-001 v1 (SUPERSEDED); 0.706	FIN-POL-001 v2 (SUPERSEDED); 0.735
Correct primary source in Top-5	Yes	Yes
SUPERSEDED in Top-5	3	3

Q2. What reasons must a creditor provide when an AI model causes an adverse credit decision?

Measure	Baseline A	Alternative B
Rank-1 result	DOC001 v1.0 (CURRENT); 0.785	DOC001 v1.0 (CURRENT); 0.810
Correct primary source in Top-5	Yes	Yes
SUPERSEDED in Top-5	0	0

Q3. How should a loan approval be escalated under the current procedure?

Measure	Baseline A	Alternative B
Rank-1 result	FIN-PROC-001 v2 (CURRENT); 0.757	FIN-PROC-001 v2 (CURRENT); 0.735
Correct primary source in Top-5	Yes	Yes
SUPERSEDED in Top-5	1	0

6. Interpretation and limitations

Question 1: Both configurations retrieve the authoritative underwriting policy **v3 CURRENT at rank 2**, but rank a SUPERSEDED policy first (A: v1; B: v2). This demonstrates why semantic similarity alone cannot establish authority. The correct document appears in Top-5 in both runs, but the first result is unsafe to treat as current without checking metadata.

Question 2: Both configurations return five CFPB chunks out of five, with a relevant passage at rank 1. A’s highest-ranked chunk includes a concrete example explaining why vague adverse-action reasons are insufficient; B retrieves more compact passages across the circular. Similarity scores differ, but higher cosine similarity is not itself proof of better answer quality.

Question 3: A retrieves three escalation-procedure chunks, one current underwriting-policy chunk and one superseded-policy chunk. B retrieves four escalation-procedure chunks and one current underwriting-policy chunk. B therefore gives more

relevant procedure coverage and avoids superseded material in this query's Top-5. The B rank-3 chunk explicitly states that escalation is a workflow action and the stored application status remains **Under Review**.

Metadata quality limitations: Some document-body statements about which policy is current conflict with the registry's authority status; authoritative version selection should rely on controlled metadata. The CFPB source URL in exported metadata includes a *utm_source* tracking parameter. This should be normalized for cleaner provenance. A procedure chunk also shows an in-document "APPROVED" status while registry metadata records CURRENT; these represent different vocabularies and should be reconciled or clearly distinguished.

7. Recommendation and next architecture step

Prefer Configuration B provisionally (800-character chunks, 100-character overlap) for this corpus: it preserved Top-5 primary-source retrieval for all three questions and improved the relevance mix for the escalation query. Configuration A remains defensible when more complete surrounding context is important. This is a small, three-question evaluation rather than a statistically robust benchmark.

For production-style retrieval, apply **CURRENT authority filtering** when the question asks for current policy, and consider document-type and access-group constraints before ranking. Evaluate filtered results separately from the unfiltered A-versus-B chunking experiment. The filtered test is not included in the six uploaded comparison outputs and is therefore not reported here as verified evidence.

8. Reproducibility / evidence inventory

Chunk exports: *baseline_A_chunks.json* (69 rows), *alternative_B_chunks.json* (144 rows). Retrieval exports: *retrieval_A_question1.json*, *retrieval_B_question1.json*, *retrieval_A_question2.json*, *retrieval_B_question2.json*, *retrieval_A_quesiton3.json*, *retreival_B_question3.json* (last two filenames retain their original spelling). All six outputs report authority_filter = null, Top-K = 5, and the same embedding model.